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Paleomagnetic Evidence for the Existence of the Geomagnetic Field 3.5 Ga Ago

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A paleomagnetic study has been made of the red dacite and pillow basalt of the Duffer Formation of the Pilbara craton, Australia. The dacite is currently the oldest dated rock unit in Australia, having a zircon U-Pb age of 3452 ± 16 Ma. Vector analysis of alternating field and thermal demagnetization results yield three components of magnetization. The high-temperature, high-coercivity hematite (H) component in the red dacite appears to be postfolding in age, as does a low-temperature, low-coercivity (L) component observed in the pillow basalt. However, an intermediate-temperature, intermediate-coercivity magnetite (M) component observed in both the pillow basalt (five sites) and the red dacite (six sites) is clearly prefolding in age and thus older than 3.0 Ga. This M component appears to be the oldest magnetization yet recorded in paleomagnetic studies and yields a paleomagnetic pole, probably original, at 43.9°N, 86.3°E (A_{95} = 7.2°). A tentative Archean apparent polar wander path for Australia suggests that the L and H components are also ancient magnetizations, probably 3.0 and 2.6 Ga old, respectively. The primary M component establishes the existence of the geomagnetic field at least 3.5 Ga ago.

INTRODUCTION

The Pilbara craton is the oldest Precambrian nucleus so far dated on the Australian continent. It consists basically of an Archean layered succession or greenstone sequence preserved within several linear and arcuate belts between large oval-shaped granite batholiths (Figure 1). An age of 3452 ± 16 Ma, using the U-Pb method on zircons from some volcanics near the base of the layered succession, is currently the oldest date observed on the Australian continent [*Pidgeon,* 1978]. The excellent exposure of the dated volcanics and their freshness prompted an attempt to obtain paleomagnetic data from an Archean sequence for the first time in Australia. A previous attempt has been made to obtain paleomagnetic data from the oldest dated rocks of Greenland [*Fahrig and Bridgewater,* 1976]. However, an original magnetization relating to the time of formation of these rocks (3700 Ma) could not be determined. The oldest preserved magnetization was apparently acquired during a later metamorphic event at about 2500 Ma.

GEOLOGY AND SAMPLING

The Archean greenstone sequence of the Pilbara craton is very diverse, containing most of the rock types found in other greenstone sequences of the world. A detailed description of the succession is given by *Hickman and Lipple* [1975]. The Warrawoona Group forms the basal, predominantly volcanic assemblage. Relatively thin chert units occur frequently interposed between ultramafic-mafic and ultramafic-felsic cycles. The upper part of the succession, the Gorge Creek Group, is predominantly a sedimentary assemblage of banded iron formations, argillaceous rocks, and sandstones with subordinate pillow basalt. The layered succession commonly exceeds 15 km in thickness and has been subjected to isoclinal folding. In the area sampled around Marble Bar (Figure 1) the beds are slightly overturned, as is clearly evidenced from the facing of the pillow basalts at the two localities. Greenschist facies metamorphism predominates in most areas. There have been two intervals of granite emplacement, a main one at around 3000 Ma [*Compston and Arriens,* 1968] and a younger event when the main granite batholiths were intruded by irregular masses of adamellite at around 2600 Ma. The folding of the Archean succession is presumed to be related to the main period of granite emplacement.

Columnar dacite from the Duffer Formation of the Warrawoona Group represents the rock type dated at 3452 Ma. It is a highly oxidized lava, and examination of polished sections shows that it is class 6 on the scale of *Wilson and Watkins* [1967]. The presence of pseudobrookite in association with rutile and hematite indicates high-temperature oxidation [*Haggerty and Lindsley,* 1969]. In view of the low-grade metamorphism associated with these rocks the oxidation almost certainly took place during initial cooling. The magnetic minerals have been oxidized very largely to hematite. In comparison with more typical basic lavas containing relatively unoxidized titanomagnetites the magnetic stability of the Duffer volcanics has been greatly enhanced. This suggests that

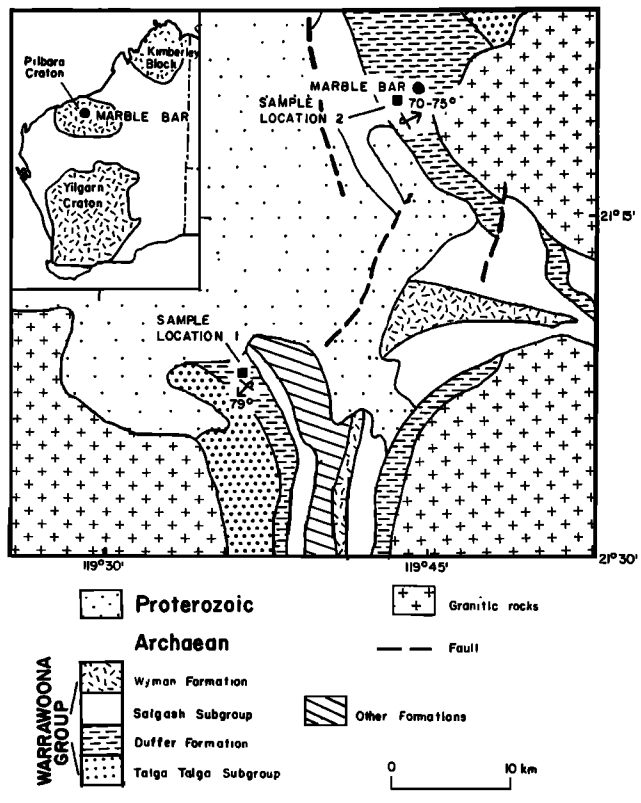


Fig. 1. Sample locality map for 3452-Ma Duffer Formation volcanics from the Pilbara craton, Australia.

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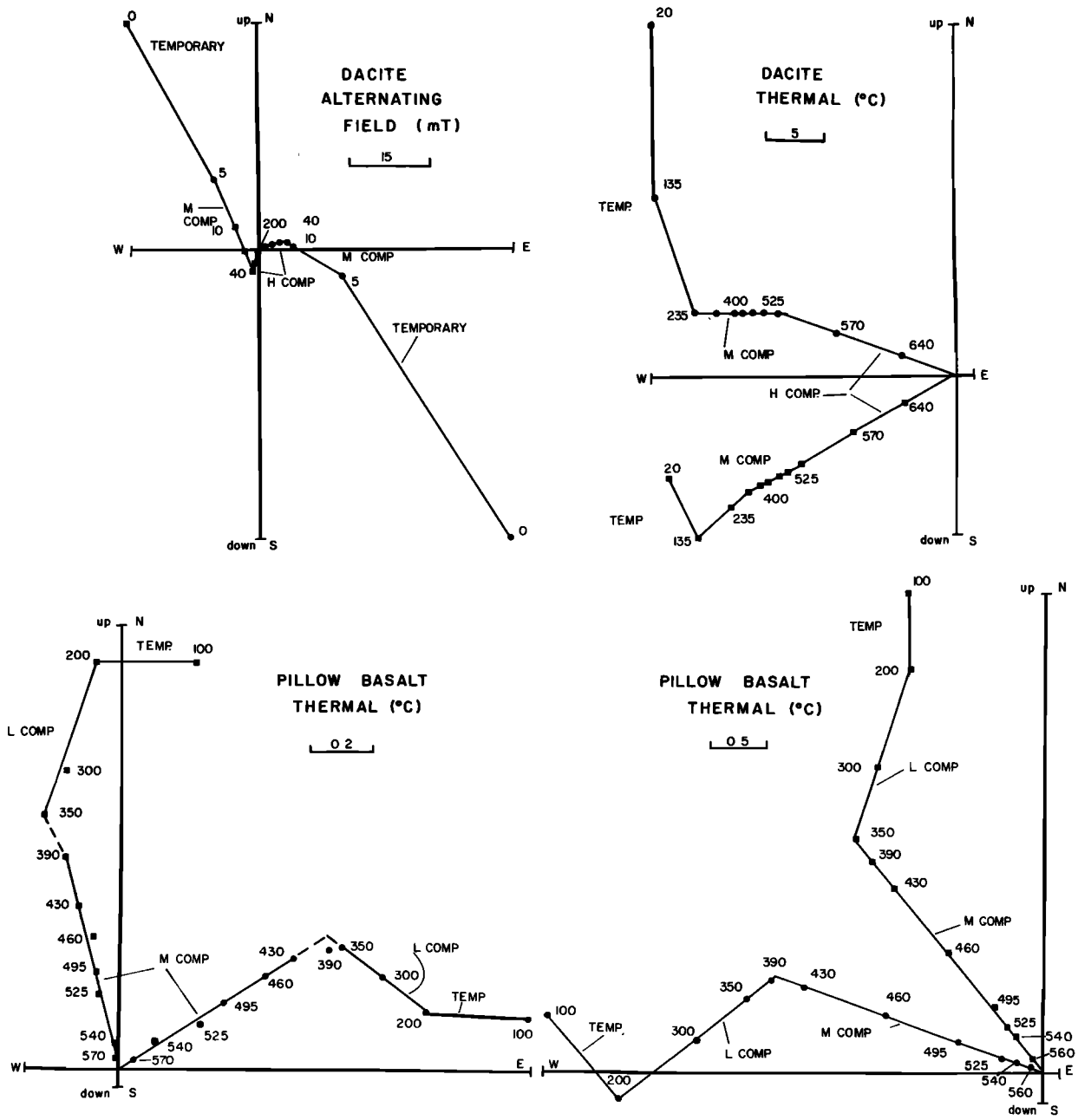


Fig. 2. Variations in declination (squares) and inclination (circles) for specimens from the Duffer Formation during demagnetization, using a modified vector diagram method of *Zijderveld* [1967]. The scale bar is given in units of mA m^{-1}.

they are ideal material for the study of the Archean of the Pilbara. Their capability of withstanding greenschist metamorphism should be optimized [*Pullaiah et al.*, 1975].

Oriented samples were collected from the columnar dacite of the Duffer Formation at the locality (location 1 in Figure 1) used by *Pidgeon* [1978] for dating these rocks. Thirty-five samples were collected, covering about 500 m in thickness. No flows can be distinguished, but they have been grouped into six horizons to represent separate points in time. At the present time the columns lie almost horizontally, the lavas being overturned corresponding to a dip of 101° toward the NE. Near Marble Bar, about 30 km away, rocks of the Duffer Formation are also overturned corresponding to dips of 105°–110° in a SW direction. Twenty-three samples were collected from pillow basalts at five separate horizons in this region. A comparison of magnetization directions between the two localities involves a marked structural correction in applying a fold test.

Paleomagnetic Results

Specimens from each sample were subjected to stepwise alternating field and thermal demagnetization, in each case involving 13 steps up to 200 mT and 640°C, respectively. The apparatus described by *McElhinny* [1966] and *McElhinny et al.* [1970] was used for these experiments, and measurements were made either on a Digico complete results magnetometer or on a Superconducting Technology two-axis cryogenic magnetometer interfaced to a Digico minicomputer. The response of typical specimens is shown on the vector diagrams of Figure 2. These are the so-called modified Zijderveld diagrams [*Zijderveld*, 1967], in which variations in declination and inclination are shown independently.

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TABLE 1. Analysis of Magnetizations Observed in the Duffer Formation

Site (Component)	N	In Situ D	In Situ I	R	Structural Correction	Bedding D	Bedding I	Pole Position Latitude, °N	Pole Position Longitude, °E
				Individual Values					
1 (M)	6	271.9	24.8	5.823	27/101	327.2	17.1	44.1	71.4
2 (M)	6	265.3	33.4	5.462	27/101	338.0	18.8	52.2	82.6
3 (M)	5	241.2	37.9	4.796	27/101	355.2	31.4	51.3	112.2
4 (M)	5	242.1	19.0	4.623	27/101	337.3	44.0	37.9	93.5
5 (M)	10	263.5	26.2	9.117	27/101	331.9	23.5	46.5	77.7
6 (M)	5	254.5	17.4	4.810	27/101	327.3	34.9	38.3	79.1
7 (M)	4	350.0	−18.1	3.679	224/105	325.7	38.0	35.9	79.3
8 (M)	4	332.3	−16.6	3.919	224/105	326.6	21.0	44.1	70.8
9 (M)	3	351.5	−26.8	2.920	219/110.5	326.8	46.1	31.7	84.9
10 (M)	3	351.8	−33.9	2.900	219/110.5	337.1	46.2	36.5	94.3
11 (M)	4	332.7	−44.1	3.832	224/105	356.2	23.5	56.3	113.0
				Means					
1–6 (M)	6*	256.5	26.9	5.856		335.9	28.6	46.5	84.7
7–11 (M)	5*	343.9	−28.2	4.873		334.9	35.6	42.2	87.0
1–11 (M)	11*	294.9	2.9	6.922	($\alpha_{95} = 8.4$)	335.5	31.8	44.6	85.8
				Individual Values					
1 (H)	5	239.7	−21.7	4.893	(27/101)	(275.5	59.6)	22.9	186.4
2 (H)	5	229.8	−20.4	4.844	(27/101)	(271.2	68.7)	31.6	181.5
3 (H)	4	217.3	−46.5	3.844	(27/101)	(218.5	53.9)	29.0	157.4
4 (H)	4	242.9	−33.1	3.745	(27/101)	(258.5	52.6)	16.9	181.8
5 (H)	5	254.5	−23.6	4.880	(27/101)	(277.3	45.8)	9.5	192.3
6 (H)	3	264.4	−26.4	2.820	(27/101)	(275.1	36.6)	0.0	194.6
				Means					
1–6 (H)	6*	243.5	−29.7	5.789	(27/101)	(263.7	54.6)	17.4	184.0
7–11 (L)	5	6.8	23.5	4.855	(222/107)	(266.2	−37.7)	55.9	131.6

Sites 1–6 are located at 21.4°S, 119.6°E; sites 7–11 are located at 21.2°S, 119.7°E. H is the high-temperature, high-coercivity (hematite) component. M is the intermediate-temperature, intermediate-coercivity (magnetite) component. L is the low-temperature, low-coercivity component. Structural corrections are given as direction of dip/amount of dip. N is the number of samples; D is the declination east of north; I is the inclination (positive down); R is the resultant of N unit vectors.

*Site means unit weight.

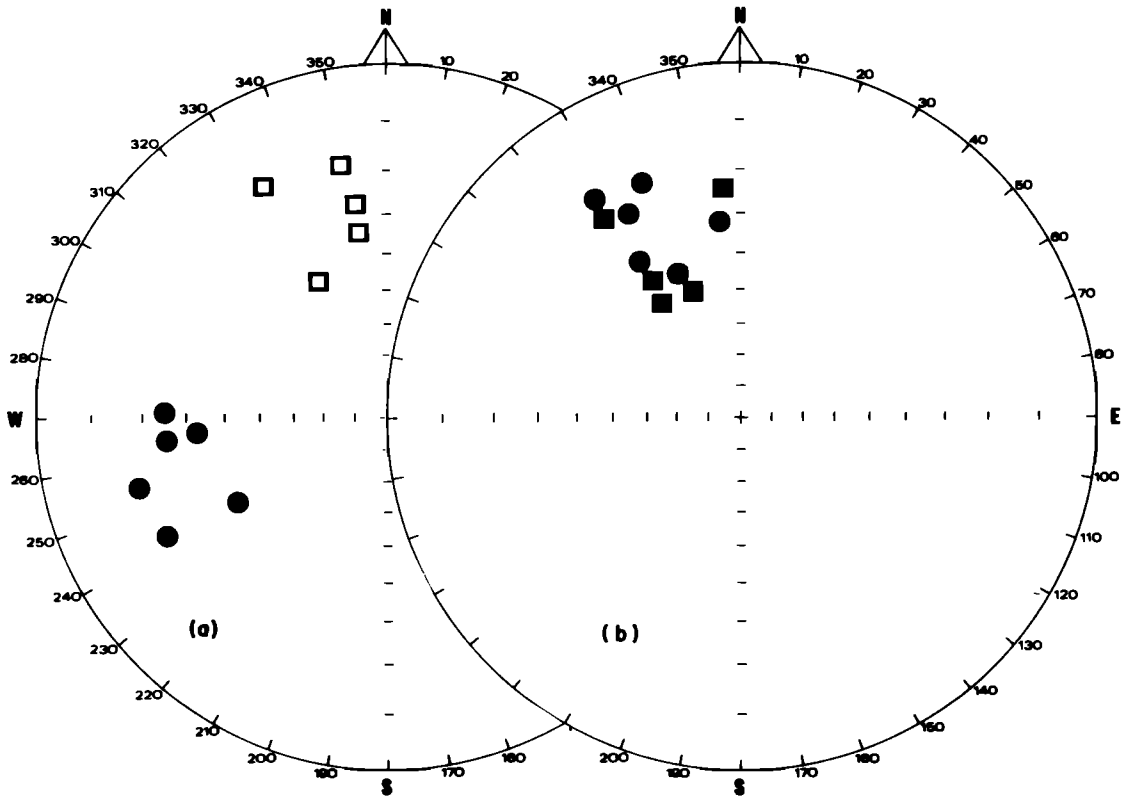


Fig. 3. Stereographic projection of M component site mean directions from the Duffer Formation (circles, dacites; squares, pillow basalts) (*a*) in situ and (*b*) after correction to the bedding plane. Solid symbols represent lower hemisphere; open symbols, upper hemisphere.

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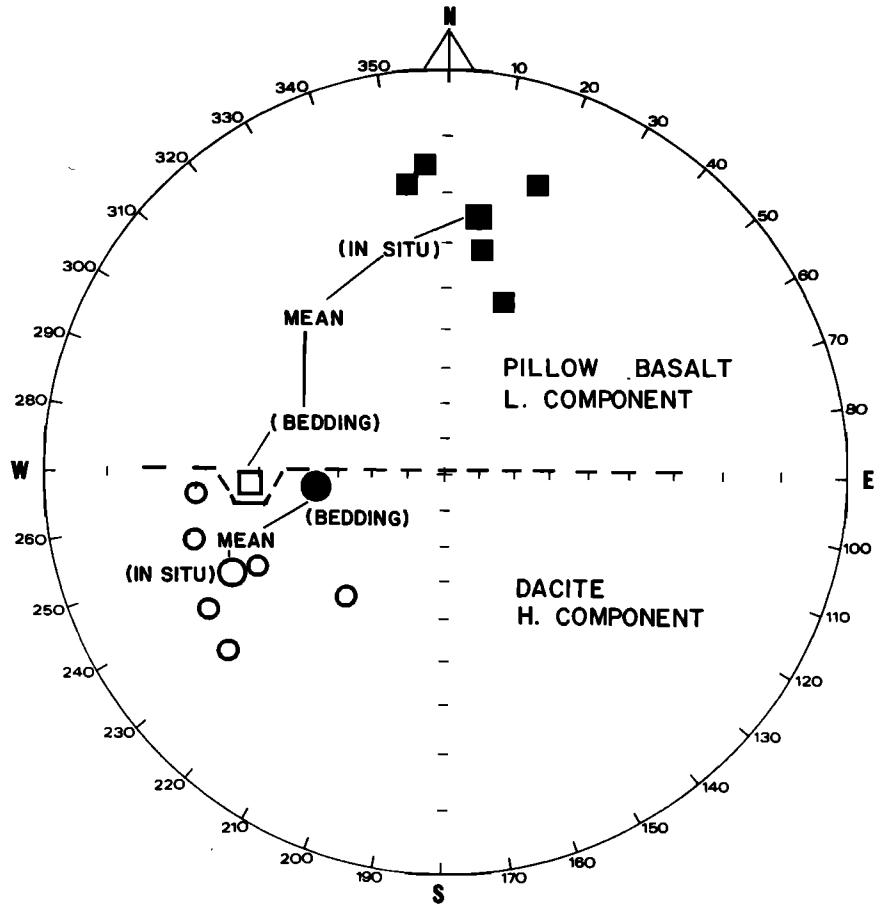


Fig. 4. *L* and *H* component site means observed during alternating field or thermal demagnetization. Stereographic projection of directions of magnetization are solid symbols on the lower hemisphere, open symbols on the lower hemisphere, and open symbols on the upper hemisphere.

Generally, three components of magnetization were observed in both the red dacite and the pillow basalt when using either alternating field or thermal techniques. All samples usually contained a temporary component acquired during storage in the present geomagnetic field. In the red dacite there is a very resistant component preserved at fields of 200 mT, and thermal demagnetization analysis shows that this is a high-temperature component with blocking temperatures above 525°C and up to at least 640°C. This high-temperature, high-coercivity component is clearly from a hematite fraction. In these rocks, thermal demagnetization is often difficult at high temperatures because of the instability of the magnetite, which tends to oxidize. There is also an intermediate-temperature (250°–500°C), intermediate-coercivity (10–40 mT) component that appears to be derived from the magnetite fraction. These components will be referred to as the H and M components, respectively (Figure 2).

TABLE 2. Mean Paleomagnetic Poles From the Duffer Formation, Pilbara Craton

Pole	*N*	Age, Ma	*K*	A_{95}	Pole Position	
					Latitude, °N	Longitude, °E
DFM	11	3452	41.0	7.2	43.9	86.3
DFL	5	~3000?	38.2	12.5	55.4	132.2
DFH	6	~2600?	22.2	14.5	18.7	182.9

N is the number of sites, *K* the precision of VGP's, and A_{95} the circle of 95% confidence about the mean pole position.

The pillow basalts did not respond coherently to alternating field treatment but showed good response under thermal demagnetization. Two examples are shown in Figure 2. Overlapping blocking temperature spectrums made it difficult to determine a low-temperature (200°–400°C) component except in a few cases. A component that is resolved up to the Curie temperature of magnetite was always discernible. The two components have been referred to as the L and M components, respectively (Figure 2).

The various components of magnetization observed in the Duffer Formation are listed in Table 1. They are listed first in situ, that is, with the rocks in their present attitude, and then after bedding, that is, after restoring the strata to the horizontal by simple rotation about their present strike. The structural corrections applied to each site are also listed in Table 1. Figure 3 shows the effect on the M components in both the dacite and the pillow basalt of restoring the strata to the horizontal. There is a marked and dramatic improvement in grouping, with directions from the two localities converging to form a single group. The *Fisher* [1953] precision parameter increases from 2.45 to 30.61 for the 11 sites, which is significant at the 99% confidence level according to the tables of *McElhinny* [1964]. This strongly suggests that these lavas have retained an original magnetization prior to folding about 3.0 Ga ago. If this is the case, this represents the oldest remanent magnetization so far observed in paleomagnetic studies.

The L and H components are listed in Table 1 and illustrated in Figure 4. After correcting the strata to the horizontal

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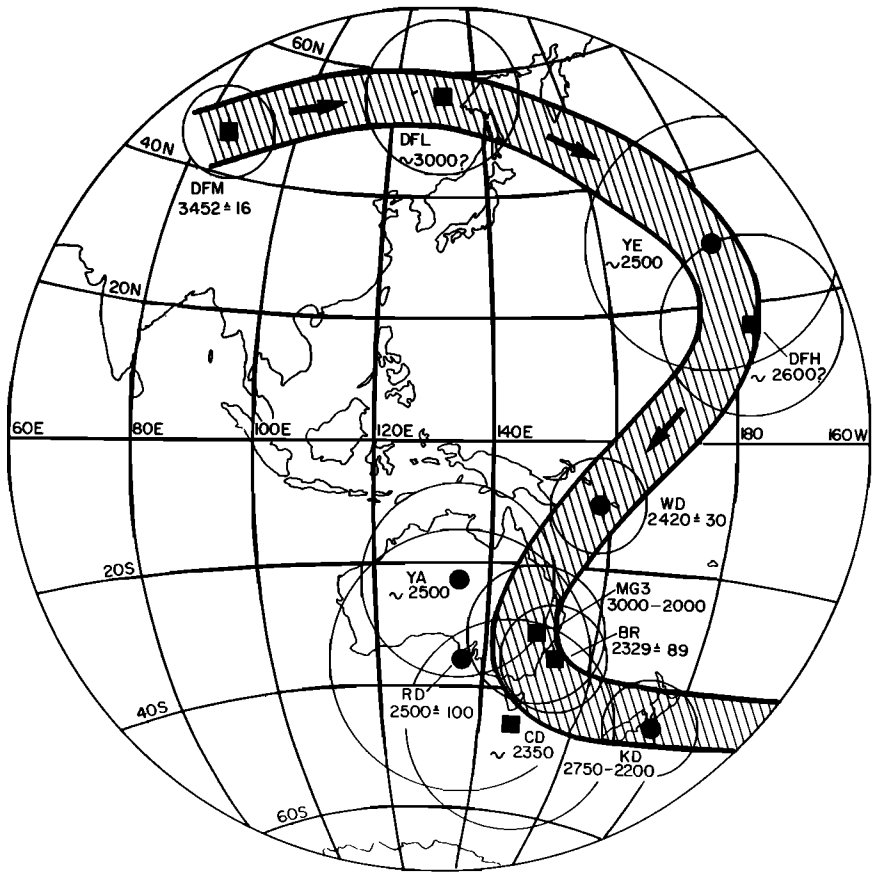


Fig. 5. Tentative Archean-Proterozoic apparent polar wander path for Australia. DFM, DFL, and DFH are the Duffer Formation poles determined in this paper. Squares are poles from the Pilbara craton; circles are from the Yilgarn craton. Symbols follow *McElhinny and Embleton* [1976] and *Embleton* [1978], best age estimates and circle of 95% confidence being given in each case.

the resulting directions do not agree either with each other or with any other known direction. Presumably, these are secondary components acquired subsequent to formation. It is surprising that the high-temperature, high-coercivity H component appears to be secondary because one would expect this to have the greatest time stability and resistance to overprinting [*Pullaiah et al.*, 1975]. However, the fact that further oxidation takes places during thermal demagnetization suggests that the formation of new hematite during some heating episode has caused the overprinting. The history of the region suggests that this took place either about 3.0 or 2.6 Ga ago.

POLE POSITIONS

Paleomagnetic pole positions calculated by combing VGP's for each of the M, L, and H components are summarized in Table 2. The L and H components are regarded as being post-folding and therefore derived from in situ data. The poles are plotted in Figure 5 along with all other pre-2.0-Ga poles from Australia. The Duffer Formation H component (DFH) magnetization gives a pole position that lies close to that of the YE dykes of the Pilbara craton, dated at ~2500 Ma [*McElhinny and Embleton*, 1976]. This strongly suggests that the DFH magnetization refers to the 2600-Ma episode in the Pilbara craton. The Duffer Formation M and L component, (DFM and DFL) magnetizations have pole positions that do not agree with any pre-2.0-Ga data or for that matter with any post-2.0-Ga results. Therefore we assume that DFM represents the primary component at 3452 Ma in view of the fold test and that DFL is a component overprinted during folding at about 3000 Ma. This view is consistent with the simplest extrapolation of the known Precambrian apparent polar wander path back to 3.5 Ga (Figure 5).

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REFERENCES

Compston, W., and P. A. Arriens, The Precambrian geology of Australia, *Can. J. Earth. Sci., 5*, 561–583, 1968.

Embleton, B. J. J., The palaeomagnetism of 2400 M.Y. old rocks from the Australian Pilbara craton and its relation to Archaean-Proterozoic tectonics, *Precambrian Res., 6*, 275–291, 1978.

Fahrig, W. F., and D. Bridgewater, Late Archean-early Proterozoic paleomagnetic pole positions from west Greenland, in *The Early History of the Earth*, edited by B. F. Windley, pp. 427–439, John Wiley, New York, 1976.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, Ser. A, 217*, 295–305, 1953.

Haggerty, S. E., and D. H. Lindsley, Stability of the pseudobrookite (Fe_2TiO_5)-ferropseudobrookite ($FeTiO_2O_5$) series, *Annu. Rep. Geophys. Lab. Carnegie Inst.*, 1968–1969, 1969.

Hickman, A. H., and S. L. Lipple, Explanatory notes on the Marble Bar 1:250,000 geological sheet, Western Australia, *West. Aust. Geol. Surv. Rec.*, 1974/20, 1975.

McElhinny, M. W., Statistical significance of the fold test in palaeomagnetism, *Geophys. J. Roy. Astron. Soc., 8*, 338–340, 1964.

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McElhinny, M. W., An improved method for demagnetizing rocks in alternating magnetic fields, *Geophys. J. Roy. Astron. Soc., 10,* 369–374, 1966.

McElhinny, M. W., and B. J. J. Embleton, Precambrian and early Palaeozoic palaeomagnetism in Australia, *Phil. Trans. Roy. Soc. London, Ser. A, 280,* 417–431, 1976.

McElhinny, M. W., G. R. Luck, and D. J. Edwards, Large volume magnetic field-free space for thermal demagnetization and other studies in palaeomagnetism, *Pure Appl. Geophys., 90,* 126–130, 1970.

Pidgeon, R. T., 3450-m.y.-old volcanics in the Archaean layered greenstone succession of the Pilbara Block, Western Australia, *Earth Planet. Sci. Lett., 37,* 421–428, 1978.

Pullaiah, G., E. Irving, K. L. Buchan, and D. J. Dunlop, Magnetization changes caused by burial and uplift, *Earth Planet. Sci. Lett., 28,* 135–143, 1975.

Wilson, R. L., and N. D. Watkins, Correlation of (1956) magnetic polarity and petrological properties in Columbia Plateau basalts, *Geophys. J. Roy. Astron. Soc., 12,* 405–424, 1967.

Zijderveld, J. D. A., A.C. demagnetization of rocks: Analysis of results, in *Methods in Palaeomagnetism,* pp. 254–286, Elsevier, New York, 1967.

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